

Facial Recognition Pipeline

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Note: in this public version of the presentation all faces are blurred. The real presentation / pipeline runs on unblurred pictures.



Facial Recognition Pipeline

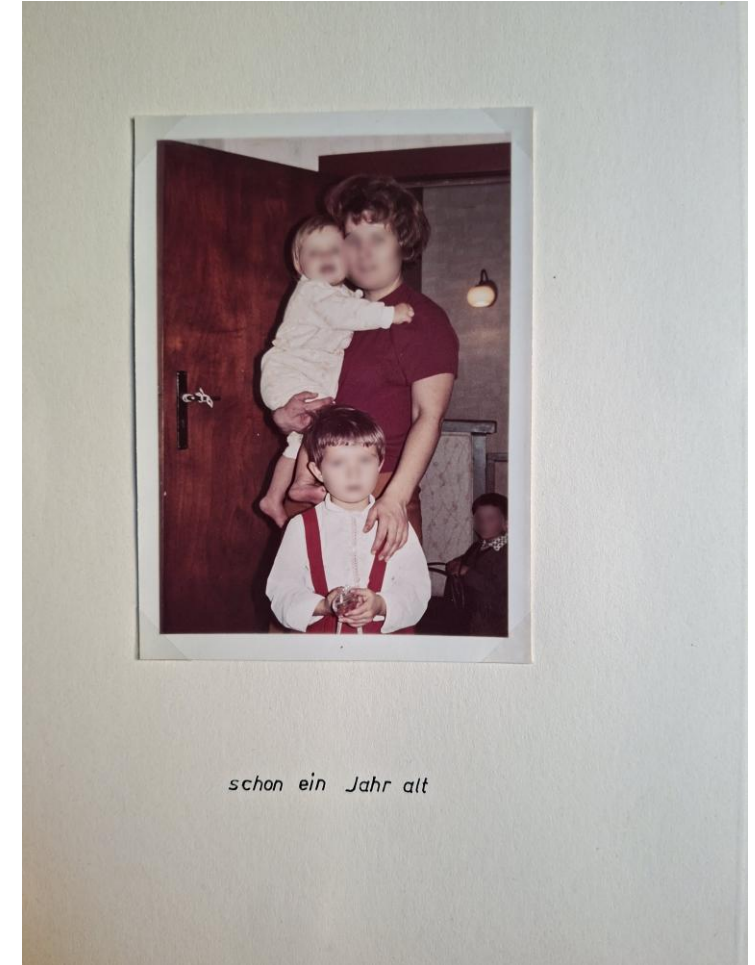
~1914



~1943



~1964

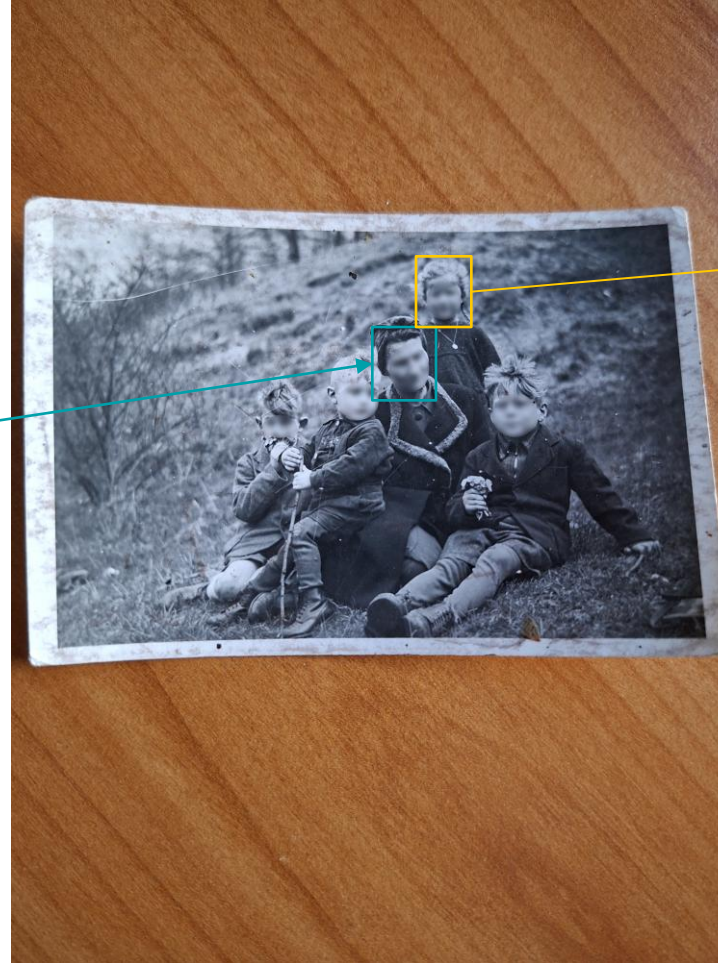


Facial Recognition Pipeline

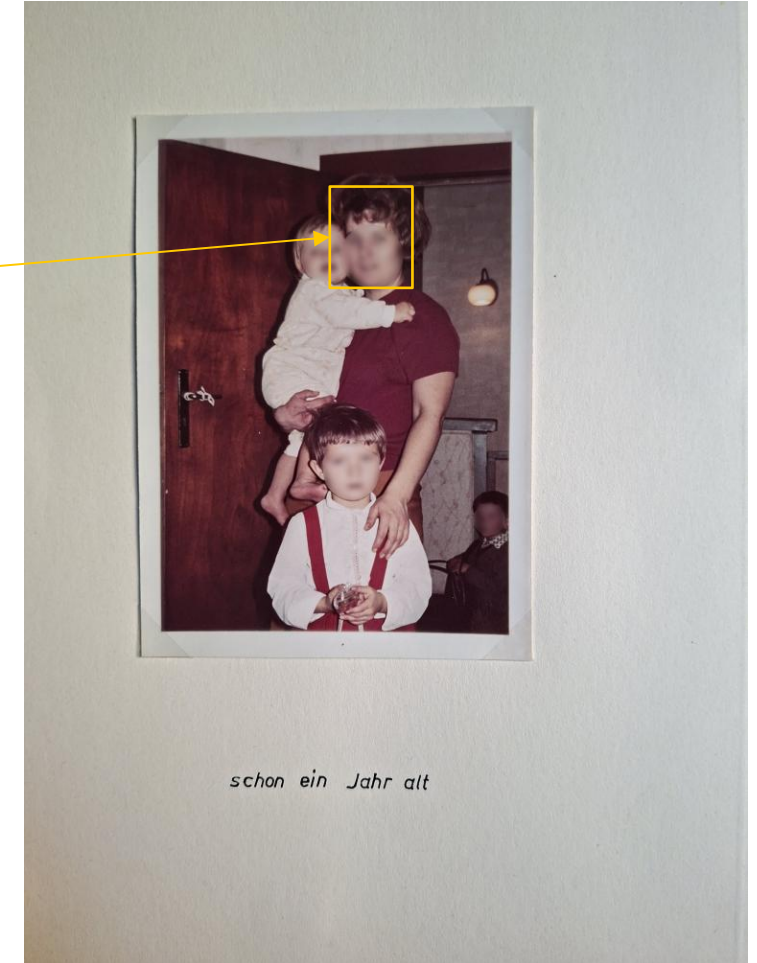
~1914



~1943



~1964



Pipeline Overview

Goal: per person gallery of pictures sorted by age

- Step 1: Download images
- Step 2: Rotate images
- Step 3: Crop to print
- Step 4: Crop to faces
- Step 5: Facial recognition
- Step 6: Age estimation

Important: all steps are idempotent

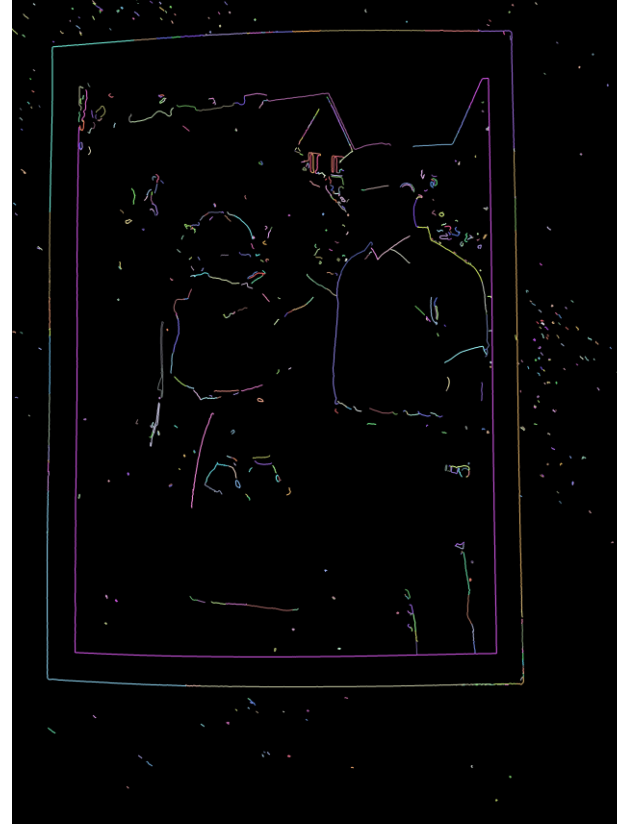


Frame Cropping: Classical Computer Vision

Original



Canny Edge Detection + Contour Tracing



contours: 712
largest: 0.0%

Morphological Closing + Contour Tracing



contours: 145
largest: 59.5%

Frame Cropping: SAM 3.1

Original



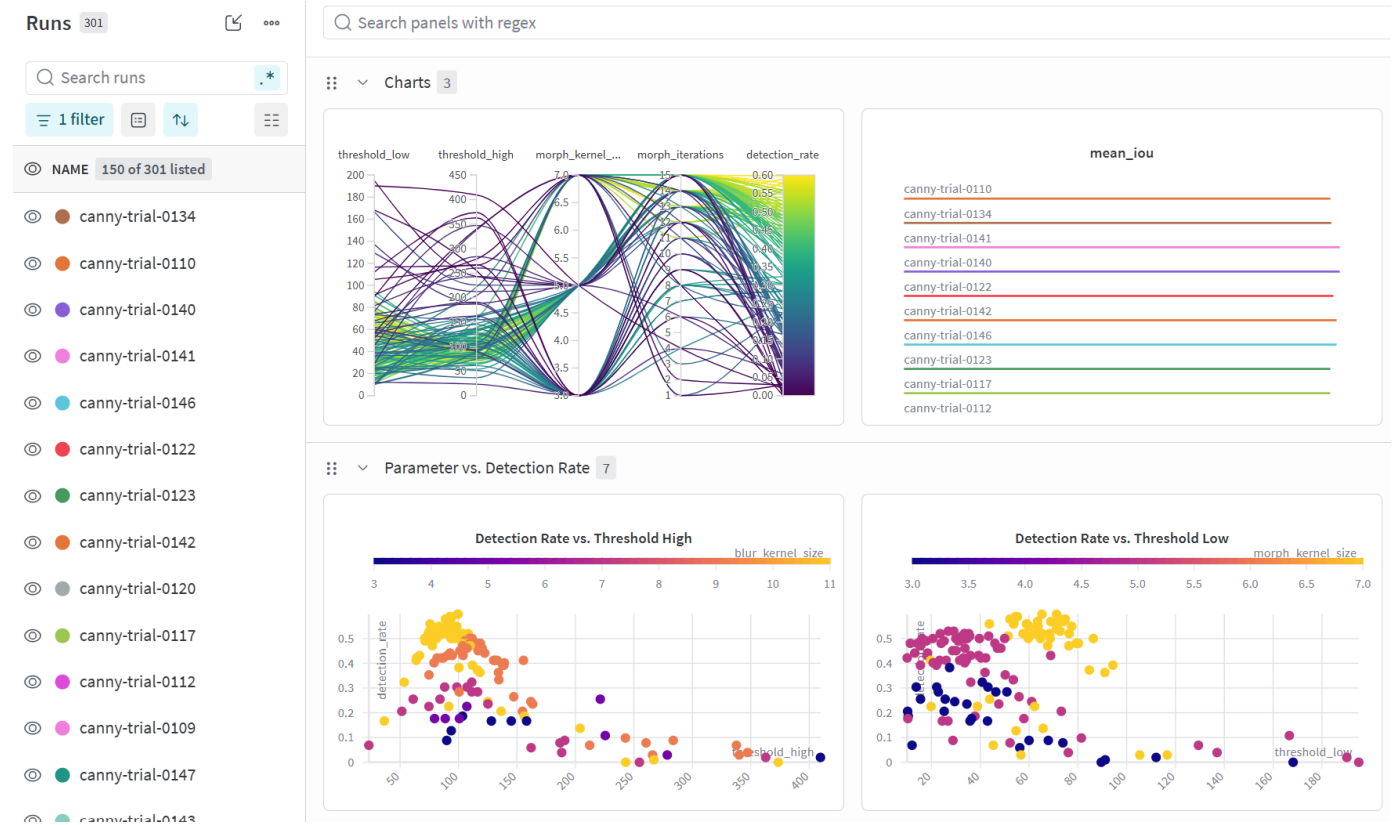
Ground Truth Annotation

- Label Studio
- Whole dataset: 1690 images
- Aim: ~200 images manually annotated
- For each picture:
 - frame bbox
 - face bboxes
 - names
 - age estimation



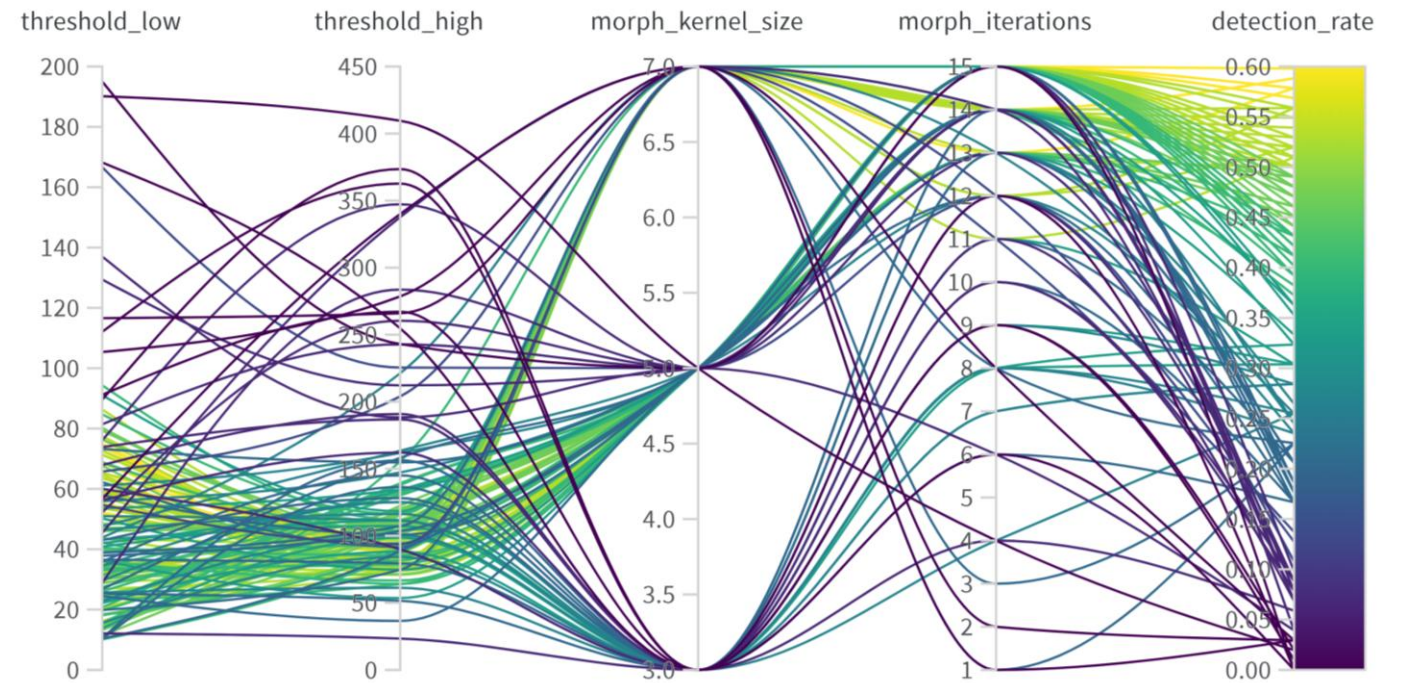
Hyperparameter Optimization

- Experiments Logging:
Weights & Biases
- Parameter Search:
Optuna TPE
(Tree-structured Parzen Estimator)



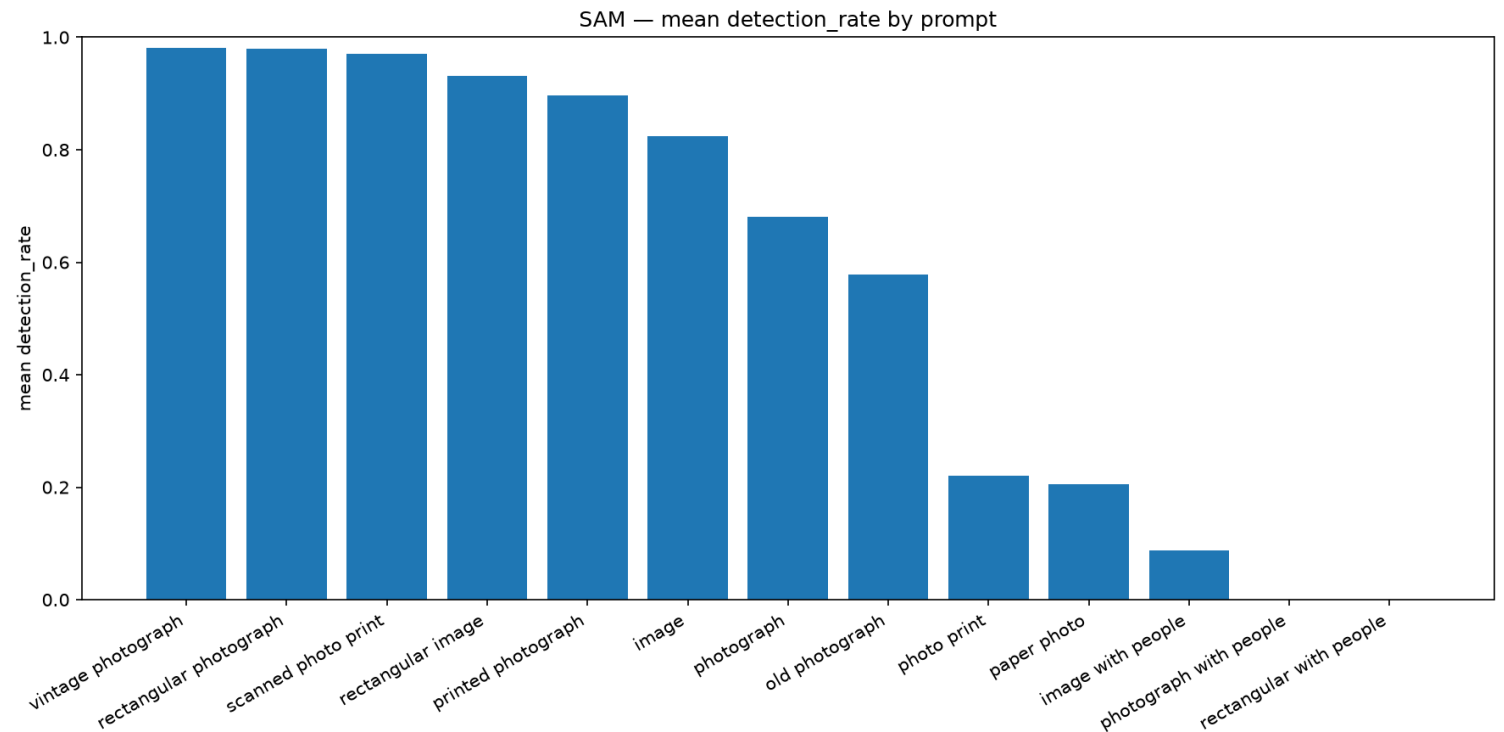
Hyperparameter Optimization – Canny Edge Detection

- 150 trials, 5 parameters
- 21m 40s
- Intel i7-1260P (local)
- Best detection rate: 59.8%



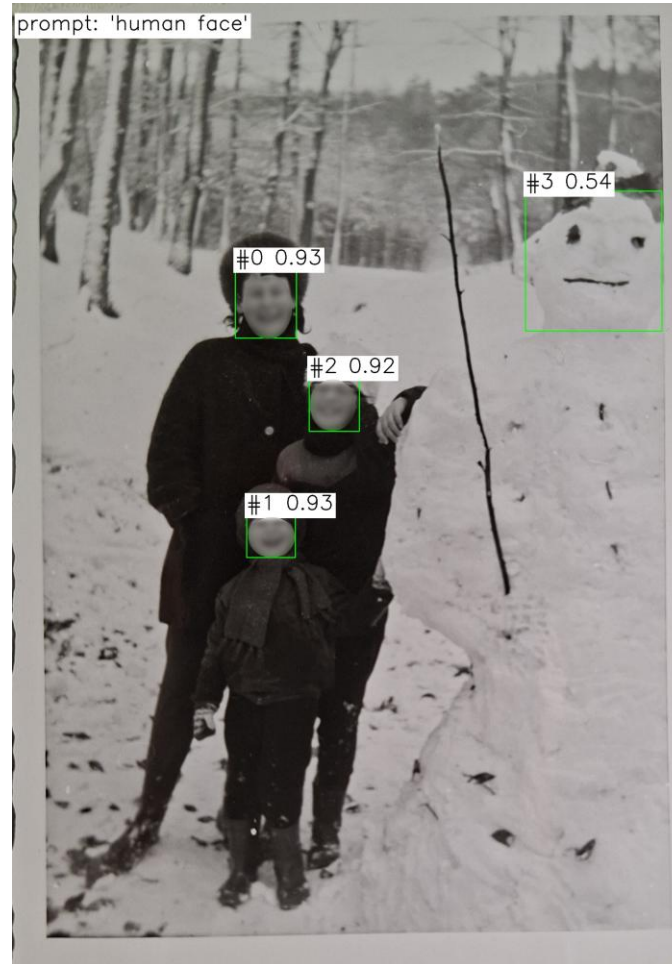
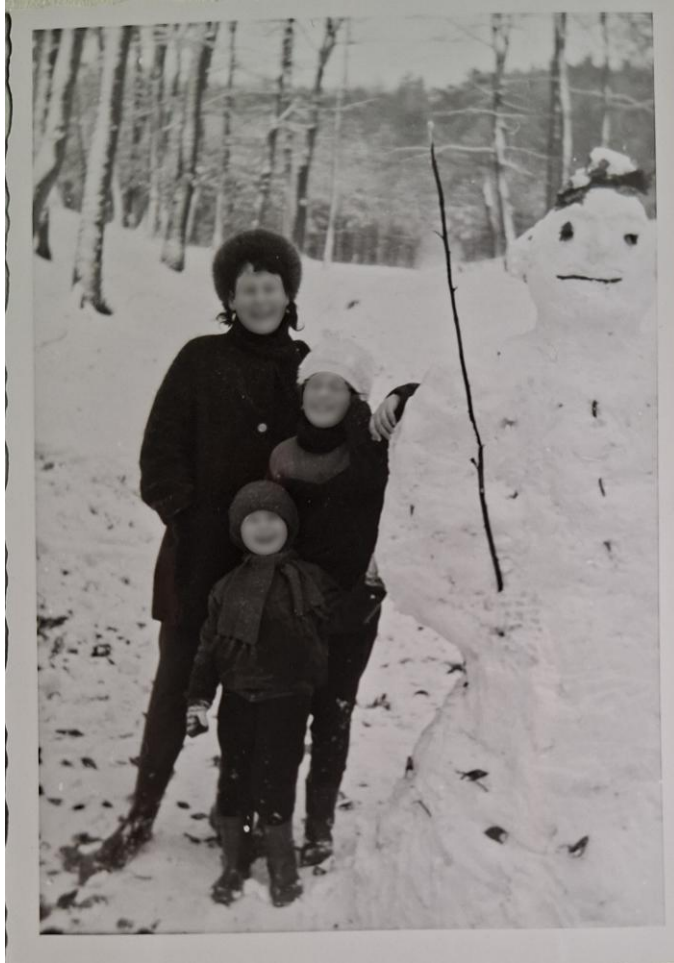
Hyperparameter Optimization – SAM 3.1

- 50 trials, 2 parameters
- 1h 24m 30s
- Tesla V100S-PCIE-32GB (BHT cluster)
- Best detection rate: 99.2%



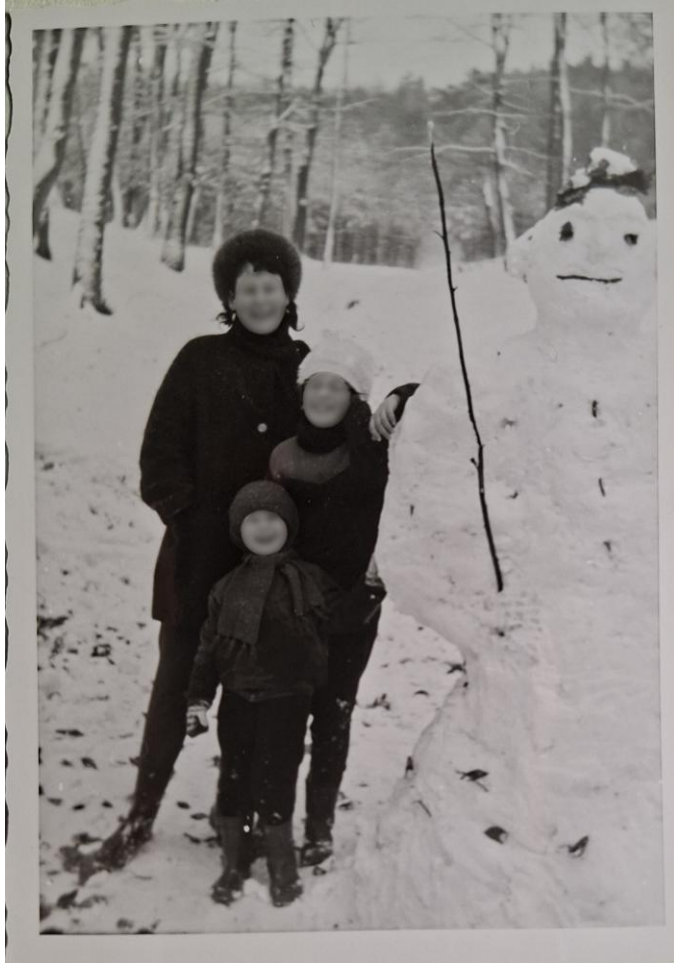
Face Cropping – SAM 3.1

Frame cropped picture



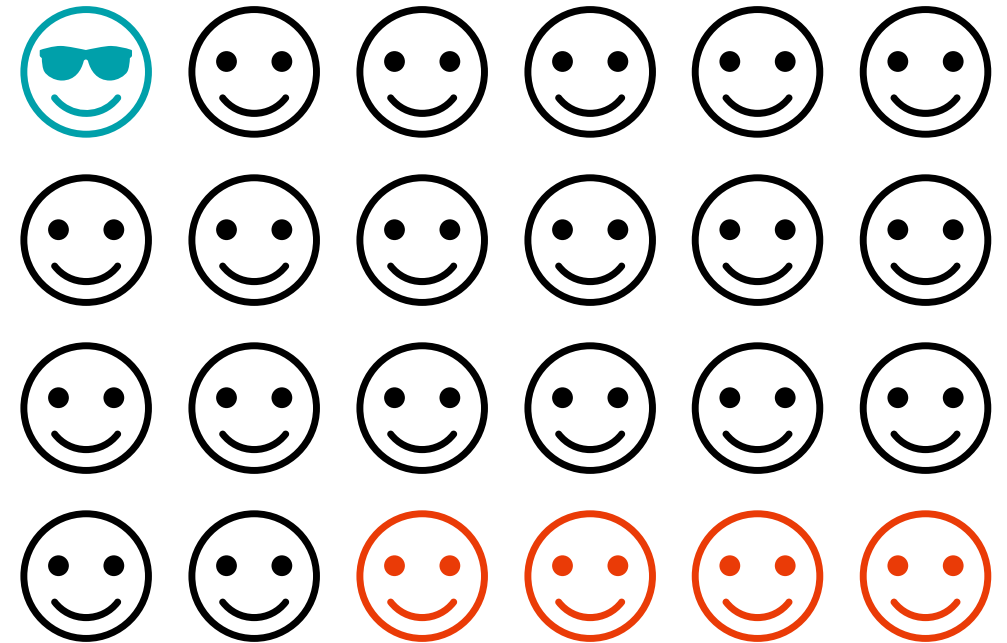
Facial Recognition – InsightFace

Frame cropped picture



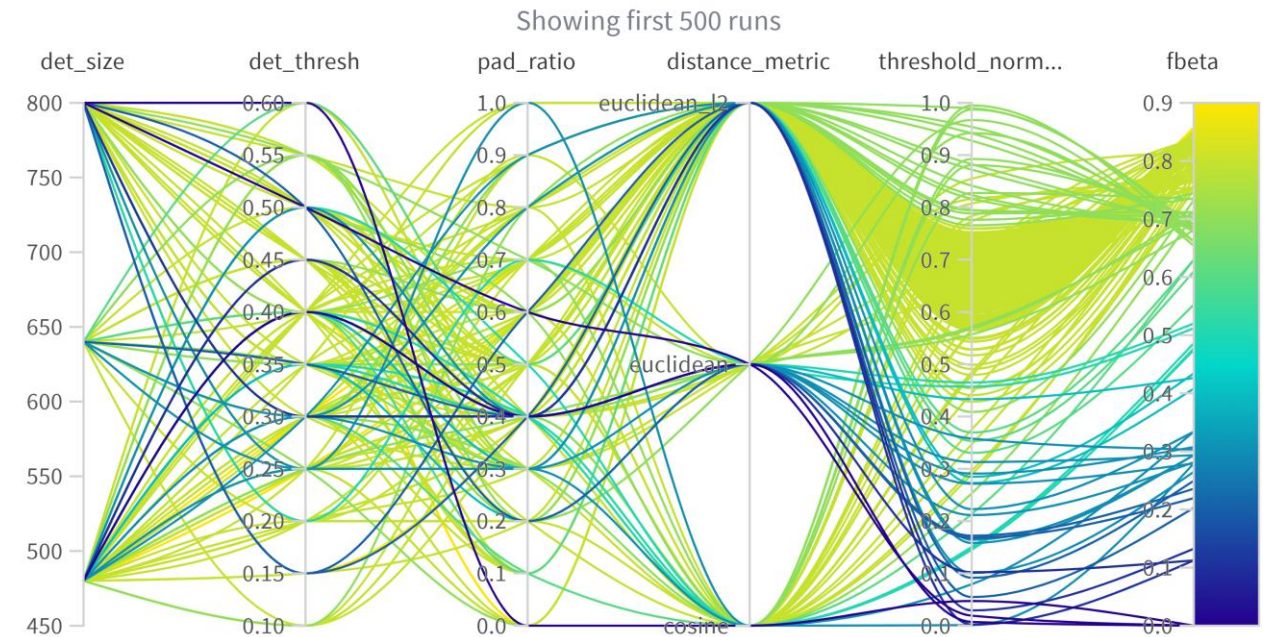
Hyperparameter Optimization – Face Recognition

- Ground truth: ~200 images, ~500 faces
- Leave One Out Cross Validation (LOOCV)
- Gallery purity is the objective
 - Optimize for Precision
 - Balance with Recall
 - Micro-average



Hyperparameter Optimization – InsightFace

- 1k trials, 6 parameters
- 2h 23m
- Tesla V100S-PCIE-32GB (BHT cluster)
- *Precision: 0.85 | Recall: 0.86 | F1: 0.85*
- Code Carbon tracking:
 - Total Energy: 0.3795 kWh
 - GPU energy: 0.1466 kWh
 - CO₂ emissions: 0.1446 kg CO₂eq



Recommender System

- “More like this” (content-based recommendation)
- Method A: *max*
(score candidate against nearest reference)

Metric	@10	@25	@50	@100
Precision	1.00	0.92	0.92	0.86
nDCG	1.00	1.00	0.99	0.98



Recommender System

- “More like this” (content-based recommendation)
- Method B: *centroid*
(score candidate against nearest reference)

Metric	@10	@25	@50	@100
Precision	1.00	1.00	1.00	1.00
nDCG	1.00	1.00	1.00	1.00



Thank you

Data Quality Metrics

Dimension	Description	Count	Of total	%
Validity	Smartphone rotation quirk	36	1,690	2.1%
Relevance	Backside of photo prints	23	1,690	1.4%
Coverage	Distinct people in reference gallery	44	-	-
Coverage	Recognizable faces	2,419	4,965	48.7%
Coverage	Unrecognizable faces	1,941	4,965	39.1%
Uniqueness	Near duplicates	249	1690	14.7%